RAJANEE SHENKAR

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PROFESSIONAL EXPERIENCE

Graduate Assistant | **Stevens Institute of Technology** Hoboken, USA | Oct 2024 – May 2026

- Developed structured administrative workflows to streamline faculty, IT, and event coordination; maintained reporting pipelines to ensure timely issue resolution and operational continuity.

Data-Focused Backend Developer | **DigitalDose** Mumbai, India | Feb 2024 – May 2024

- Built NLP-based intent classification system using Naive Bayes and scikit-learn for automated query routing; integrated TensorFlow Lite ML pipelines for real-time client-side inference.
- Developed and maintained REST APIs and backend data pipelines for a production CRM platform; designed MongoDB schemas with structured validation and role-based access control via JWT.
- Authored Postman test suites verifying authentication, CRUD operations, and data integrity.

Data Engineer | **Yibe** Pune, India | Jul 2022 – Dec 2022

- Engineered microservices for authentication and payment data processing; integrated Firebase Authentication and Firestore for structured session and transaction data management.
- Wrote Python scripts for automated data extraction and regression testing, identifying edge-case failures early and reducing post-release bugs by 40%.

Application Support Analyst | **Iraje Software** Mumbai, India | Dec 2021 – Mar 2022

- Developed optimized SQL queries for an enterprise Privileged Access Management system, improving data retrieval efficiency and supporting structured data management for enterprise clients.
- Diagnosed and resolved application-level issues using data structures knowledge to optimize query response times and bug resolution cycles.

TECHNICAL SKILLS

Languages: C++, Python, R, SQL

LLM & AI Engineering: OpenAI API, RAG, Multi-turn Context, Prompt Engineering, Vector Search
AI Frameworks & Tools: LangChain, HuggingFace, Chroma, FastMCP, tiktoken, PyMuPDF, BeautifulSoup, TensorFlow, PyTorch, Scikit-Learn
APIs & Data: Kaiko Market & Reference Data API, USPTO PatentsView API, Twitter API, Pandas, NumPy
Cloud & Infrastructure: AWS, Azure, Snowflake, Databricks, MongoDB, Firebase, Git, Hadoop,

PROJECTS

Patent RAG Agent | *Python, OpenAI Responses API, Function Calling, Vector Search, USPTO API*

- Built a multi-tool AI agent using the OpenAI Responses API with function calling, implementing three retrieval tools (vector search, google_patent_fetch, uspto_search) over a corpus of 10 patents and 1,679 chunks. Engineered a patent ETL pipeline using tiktoken for chunking and OpenAI embeddings for vectorization.

Kaiko MCP Agent | *Python, FastMCP, Kaiko API, OpenAI API, Async*

- Built a production-style MCP server using FastMCP exposing Kaiko reference data endpoints as callable tools, and an async agent client routing queries through OpenAI function calling. Extended to download OHLCV and raw trade data with paginated fetching, CSV export, and a multi-turn agent loop.

Audio Classification & Speech Transcription | *Python, OpenAI Whisper, scikit-learn*

- Built an audio classification and speech-to-text transcription system using OpenAI Whisper; trained a gender classifier with scikit-learn, exported as a production-ready .pkl model, with transcription outputs validated against ground truth.

EDUCATION

Stevens Institute of Technology - M.S., Data Science Hoboken, NJ | Sep 2024 – May 2026
Coursework: Prompt Engineering, Applied ML, Deep Learning, Time Series Analysis, Data Visualization, Financial DB Design, Statistical Methods

MIT-WPU - B.Tech, Computer Science & Engineering Pune, India | Aug 2019 – Jun 2023
Coursework: Artificial Intelligence, Big Data Analytics, Data Structures & Algorithms, DBMS, Distributed Computing, Microprocessor and Microcontroller, Operating Systems, Information Security

PUBLICATION

Advanced Generation of Private and Community Crowdfunding Platform for Investors | *European Chemical Bulletin*
Published research on a blockchain-based crowdfunding platform leveraging Ethereum, Solidity, Smart Contracts, and Web3.js for decentralization and transparency.